barely revealed the existence of the lobes (two rectangles in Figure 9.3d), and with Ryle's 1969 contour map (Figure 9.3d). [courtesy NRAO/AUI.]

These VLBI pictures showed, in the early 1980s, that the jets extend right into the innermost few light-years of the core of a galaxy or quasar—the very region in which resides, in the case of some quasars such as 3C273, a brilliantly luminous, light-emitting object no larger than a light-month in size. Presumably the central engine is inside the light-emitting object, and it is powering not only that object, but also the jets, which then feed the radio lobes.

The jets gave yet another clue to the nature of the central engine. Some jets were absolutely straight over distances of a million light-years or more. If the source of such jets were turning, then, like a rotating water nozzle on a sprinkler, it would produce bent jets. The observed jets' straightness thus meant that the central engine had been firing its jets in precisely the same direction for a very long time. How long? Since the jets' gas cannot move faster than the speed of light, and since some straight jets were longer than a million light-years, the firing direction must -have been steady for more than a million years. To achieve such steadiness, the engine's "nozzles," which eject the jets, must be attached to a superbly steady object—a long-lived *gyroscope* of some sort. (Recall that a gyroscope is a rapidly spinning object that holds the direction of its spin axis steadily fixed over a very long time. Such gyroscopes are key components of inertial navigation systems for airplanes and missiles.)

Of the dozens of ideas that had been proposed by the early 1980s to explain the central engine, only one entailed a superb gyroscope with a long life, a size less than a light-month, and an ability to generate powerful jets. That unique idea was a gigantic, spinning black hole.

Gigantic Black Holes

The idea that gigantic black holes might power quasars and radio galaxies was conceived by Edwin Salpeter and Yakov Borisovich Zel'dovich in 1964 (the first year of the golden age—Chapter 7). This idea was an obvious application of the Salpeter–Zel'dovich discovery that gas streams, falling toward a black hole, should collide and radiate (see Figure 8.4).